A Person Obesity Category Predicted from Eating Habits and Lifestyle rather than BMI

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August 29, 2026

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Introduction

Many individuals every year are told by a doctor that their weight puts them at risk for obesity, but can a person's obesity category be predicted from their eating habits and lifestyle rather than from direct BMI? Using this public dataset, I analyzed whether there are unclear associations with eating and activity habits that always occur with higher obesity categories like consumption of high-caloric food together with low physical activity is associated with a specific obesity level. I used classification learning as the main approach, since the target attribute (NObeyesdad) is a separate class with seven possible values and the goal is to predict it from other attributes. I used three classifiers: J48, Naïve Bayes and Random Forest. I also used Apriori because it will look for combinations of lifestyle habits that tend to occur together often and reliably. In result Random Forest was the strongest performer, coming out with correctly classifying accuracy of 88% using Lifestyle attributes and 96% using All attributes. The reasoning behind the two results will be explained further in the report.

Description

The dataset I chose to analyze is “Estimation of Obesity Levels Based on Eating Habits and Physical Condition,” located on the UCI Machine Learning Repository. It contains over 2,100 records describing the eating habits, physical activity, and demographics of individuals from Mexico, Peru, and Colombia.

https://archive.ics.uci.edu/dataset/544/estimation+of+obesity+levels+based+on+eating+habits+a nd+physical+condition

These are the key attributes:

Attribute	Type	Description
Gender	Categorical	Female / Male
Age	Numeric	Age
Height	Numeric	Height (meters)
Weight	Numeric	Weight (kilos)
family_history_with_over weight	Binary (y/n)	Has a family member suffered/suffers from overweight?
FAVC	Binary (y/n)	Do you eat high caloric food frequently?
FCVC	Numeric (1-3)	Do you usually eat vegetables in your meals?
NCP	Numeric	How many main meals do you have daily?
CAEC	Categorical	Do you eat any food between meals? (No / Sometimes / Frequently / Always)
SMOKE	Binary(y/n)	Do you smoke?
CH2O	Numeric	How much water do you drink daily?
SCC	Binary (y/n)	Do you monitor the calories you eat daily?
FAF	Numeric	How often do you have physical activity?
TUE	Numeric	How much time do you use technological devices such as cell phone, videogames, television, computer and others?
CALC	Categorical	How often do you drink alcohol?
MTRANS	Categorical	Which transportation do you usually use?
NObeyesdad (class) | Target	Categorical (7 levels)	Obesity level

Preparation

Upon receiving the dataset as a CSV file, it was converted to an ARFF file in Weka. It was also cleaned of any duplicate records which came to be 24 records removed, left with 2087 records. These 24 duplicated records were developed in the dataset SMOTE data augmentation process, which creates modified copies of data.

The records were also randomized (with a fixed seed of 42) before splitting the data, since they were originally connected groups by obesity categories. I changed it to be randomly ordered so the train and test split can include every obesity category there is. After saving the clean dataset as an ARFF file, it was split into training (80%) (1670 instances) and test (20%) (417 instances) sets using Weka. The Remove Percentage filter was used twice with the same random seed so the resulting files line up to match the training and test set with no instances shared between them.

There were 2 versions of the attribute set saved for the classification portion, labeled in the tables below as Lifestyle (main.arff) and All (clean.arff).The All version held all attributes including the height and weight that were used as another point of comparison. The Lifestyle version excluded weight and height, since the obesity category in the dataset is based on directly from the BMI (body mass index) which is weight divided by height squared. Removing these attributes would make the classification testing of whether eating and lifestyle habits alone carry predictive signals more significant.

For the Apriori association since it requires nominal categorical attributes, 6 attributes that were numeric (age, veg consumption, number of main meals, daily water intake, physical activity and technology use) were grouped into 3 categories using Weka's Discretize filter.

Analysis

In this project, 3 classification algorithms were used on each train/test split described previously and an association ruled algorithm were used on the full 2087 instance dataset.

J48 – builds a C4.5 decision tree by repeatedly choosing the attribute that best separates the training examples into their classes, based on information gain, and continues splitting until the remaining groups are pure. The result is a set of readable if-then rules that can be traced from the root of the tree to a leaf (Witten, Frank, Pal & Hall, 2016).

Naïve Bayes - estimates the probability of each class by combining each probability input of every attribute value, under the simplifying assumption that the attributes are independent of one another given the class. It is fast to train, but often can be inaccurate when attributes are correlated (GeeksforGeeks, Feb 2026).

Random Forest - builds many individual decision trees, each trained on a random portion of the data and attributes. Then combines their predictions by the majority. Balancing out across many trees improving accuracy and makes the model less sensitive to overfitting rather than a single decision tree(GeeksforGeeks, May 2026)..

Apriori – an algorithm that searches for combinations of attribute values that occur together more often than would be expected by chance. Each rule it produces has a support value (how frequently the pattern appears in the data) and a confidence value (how often the rule holds true when its condition is met). Every rule's conclusion is a specific obesity category, which lets the algorithm eating and lifestyle combinations that are likely to coexist with obesity levels(GeeksforGeeks, May 2026)..

Results

The three classifiers were evaluated on a train/test split of the cleaned dataset. Each classifier was run twice, once using the Lifestyle attributes and the other with All attributes including the height and weight to estimate how much those 2 attributes influence the model's ability to predict obesity category.

Summary of Model Performance

Lifestyle Attributes = no Weight/Height | All Attributes = includes Weight/Height

Dataset Variations	Type	Correctly Classified	Kappa	MAE	RMSE	RAE	RRSE	Total
Lifestyle	J48	327 (78.4173%)	0.7475	0.071	0.2328	29.0036%	66.5426%	417
Lifestyle	Naive Bayes	227 (54.4365%)	0.4656	0.1498	0.2912	61.2044%	83.2296%	417
Lifestyle	Random Forest	368 (88.2494%)	0.8625	0.0834	0.1745	34.0771%	49.8814%	417
All	J48	386 (92.5659%)	0.913	0.0231	0.1417	9.4316%	40.4934%	417
All	Naive Bayes	288 (69.0647%)	0.6377	0.1064	0.2507	43.4737%	71.6466%	417
All	Random Forest	404 (96.8825%)	0.9635	0.0433	0.1096	17.7115%	31.31%	417

Based on the results, Random Forest produced the strongest results for both variants. Then J48 to come in second in both as well. Naïve Bayes didn't do so well likely because it assumes attributes are independent which isn't the best for correlated lifestyle attributes like what you eat and physical activity. Also, you see how Height and Weight do increase every classifiers accuracy like Random Forest going from 88% to 96% accuracy. In all, this suggests that lifestyle attributes can predict the obesity category pretty well, but it doesn't fully substitute the body measurement data.

Detailed Accuracy by Class

Model	Class	TP Rate	FP Rate	Precision	Recall	F1	MCC	ROC	PRC
Lifestyle J48	Normal	0.607	0.058	0.618	0.607	0.613	0.553	0.795	0.449
Lifestyle J48	Over I	0.717	0.058	0.644	0.717	0.679	0.630	0.858	0.627
Lifestyle J48	Over II	0.656	0.028	0.800	0.656	0.721	0.683	0.842	0.655
Lifestyle J48	Obese I	0.754	0.051	0.719	0.754	0.736	0.690	0.910	0.707
Lifestyle J48	Insuff.	0.889	0.014	0.906	0.889	0.897	0.882	0.932	0.804
Lifestyle J48	Obese II	0.804	0.030	0.804	0.804	0.804	0.773	0.913	0.668
Lifestyle J48	Obese III	1.000	0.012	0.950	1.000	0.974	0.969	0.996	0.962
Lifestyle J48	**Avg.**	**0.784**	**0.035**	**0.785**	**0.784**	**0.783**	**0.750**	**0.897**	**0.708**
Lifestyle Naive	Normal	0.232	0.025	0.591	0.232	0.333	0.316	0.804	0.423
Lifestyle Naive	Over I	0.245	0.041	0.464	0.245	0.321	0.272	0.800	0.375
Lifestyle Naive	Over II	0.164	0.037	0.435	0.164	0.238	0.197	0.810	0.389
Lifestyle Naive	Obese I	0.557	0.160	0.374	0.557	0.447	0.340	0.809	0.375
Lifestyle Naive	Insuff.	0.648	0.085	0.530	0.648	0.583	0.518	0.898	0.664
Lifestyle Naive	Obese II	0.821	0.125	0.505	0.821	0.626	0.575	0.919	0.679
Lifestyle Naive	Obese III	1.000	0.059	0.792	1.000	0.884	0.863	1.000	1.000
Lifestyle Naive	**Avg.**	**0.544**	**0.076**	**0.537**	**0.544**	**0.506**	**0.457**	**0.868**	**0.575**
Lifestyle Random Forest	Normal	0.714	0.042	0.727	0.714	0.721	0.678	0.961	0.805
Lifestyle Random Forest	Over I	0.830	0.016	0.880	0.830	0.854	0.834	0.963	0.891

Lifestyle Random Forest	Over II	0.770	0.017	0.887	0.770	0.825	0.800	0.985	0.940
Lifestyle Random Forest	Obese I	0.934	0.031	0.838	0.934	0.884	0.864	0.989	0.952
Lifestyle Random Forest	Insuff.	0.944	0.006	0.962	0.944	0.953	0.946	0.998	0.990
Lifestyle Random Forest	Obese II	0.946	0.019	0.883	0.946	0.914	0.901	0.994	0.978
Lifestyle Random Forest	Obese III	1.000	0.006	0.974	1.000	0.987	0.984	1.000	1.000
Lifestyle Random Forest	**Avg.**	**0.882**	**0.019**	**0.883**	**0.882**	**0.881**	**0.863**	**0.985**	**0.940**
All J48	Normal	0.911	0.025	0.850	0.911	0.879	0.861	0.965	0.797
All J48	Over I	0.849	0.005	0.957	0.849	0.900	0.888	0.932	0.849
All J48	Over II	0.869	0.008	0.946	0.869	0.906	0.892	0.963	0.886
All J48	Obese I	0.967	0.031	0.843	0.967	0.901	0.885	0.982	0.840
All J48	Insuff.	0.926	0.008	0.943	0.926	0.935	0.925	0.979	0.952
All J48	Obese II	0.929	0.006	0.963	0.929	0.945	0.937	0.960	0.909
All J48	Obese III	1.000	0.003	0.987	1.000	0.993	0.992	0.999	0.987
All J48	**Avg.**	**0.926**	**0.012**	**0.929**	**0.926**	**0.926**	**0.915**	**0.970**	**0.893**
All Naive Bayes	Normal	0.482	0.047	0.614	0.482	0.540	0.483	0.922	0.608
All Naive Bayes	Over I	0.547	0.055	0.592	0.547	0.569	0.509	0.923	0.623
All Naive Bayes	Over II	0.492	0.067	0.556	0.492	0.522	0.447	0.889	0.538
All Naive Bayes	Obese I	0.475	0.084	0.492	0.475	0.483	0.397	0.860	0.476
All Naive Bayes	Insuff	0.907	0.036	0.790	0.907	0.845	0.823	0.983	0.914
All Naive Bayes	Obese II	0.857	0.036	0.787	0.857	0.821	0.792	0.978	0.909
All Naive Bayes	Obese III	1.000	0.035	0.864	1.000	0.927	0.913	1.000	1.000
All Naive Bayes	**Avg.**	**0.691**	**0.051**	**0.676**	**0.691**	**0.680**	**0.632**	**0.938**	**0.732**
All Random Forest	Normal	0.946	0.014	0.914	0.946	0.930	0.919	0.993	0.951
All Random Forest	Over I	0.906	0.008	0.941	0.906	0.923	0.912	0.995	0.974
All Random Forest	Over II	0.951	0.006	0.967	0.951	0.959	0.952	0.998	0.993
All Random Forest	Obese I	1.000	0.006	0.968	1.000	0.984	0.981	1.000	1.000
All Random Forest	Insuff	0.981	0.000	1.000	0.981	0.991	0.989	1.000	0.999
All Random Forest	Obese II	0.982	0.000	1.000	0.982	0.991	0.990	1.000	0.999
All Random Forest	Obese III	1.000	0.003	0.987	1.000	0.993	0.992	1.000	1.000
All Random Forest	**Avg.**	**0.969**	**0.005**	**0.969**	**0.969**	**0.969**	**0.964**	**0.998**	**0.989**

For the Confusion Matrix, it shows how a classifier's predictions compare to the true labels for every test instance (referred to as N, which equals 417 in each case). Each row is the actual obesity category and each column is the category the model predicted. They were run once on Lifestyle attributes only and then once with All attributes, matching the rows in the Summary of Model Performance and Detailed Accuracy tables above.

Lifestyle Attributes

J48 — Confusion Matrix

Actual ↓ / Predicted →	Normal	Over I	Over II	Obese I	Insufficient Weight	Obese II	Obese III
Normal	34	7	4	4	3	2	2
Over I	8	38	1	3	2	1	0
Over II	5	2	40	8	0	6	0
Obese I	2	8	2	46	0	1	2
Insufficient Weight	4	1	0	0	48	1	0
Obese II	2	3	3	3	0	45	0
Obese III	0	0	0	0	0	0	76

Naive Bayes — Confusion Matrix

Actual ↓ / Predicted →	Normal	Over I	Over II	Obese I	Insufficient Weight	Obese II	Obese III
Normal	13	7	3	6	17	2	8
Over I	4	13	6	13	5	12	0
Over II	2	3	10	25	2	16	3
Obese I	2	2	1	34	7	11	4
Insufficient Weight	1	1	2	8	35	4	3
Obese II	0	2	1	5	0	46	2
Obese III	0	0	0	0	0	0	76

Random Forest — Confusion Matrix

Actual ↓ / Predicted →	Normal	Over I	Over II	Obese I	Insufficient Weight	Obese II	Obese III
Normal	40	3	4	5	2	2	0
Over I	5	44	1	2	0	1	0
Over II	5	1	47	4	0	4	0
Obese I	2	1	0	57	0	0	1
Insufficient Weight	2	1	0	0	51	0	0
Obese II	1	0	1	0	0	53	1
Obese III	0	0	0	0	0	0	76

Full Attributes

J48 — Confusion Matrix

Actual ↓ / Predicted →	Normal	Over I	Over II	Obese I	Insufficient Weight	Obese II	Obese III
Normal	51	2	0	0	3	0	0
Over I	5	45	3	0	0	0	0
Over II	0	0	53	8	0	0	0
Obese I	0	0	0	59	0	2	0
Insufficient Weight	4	0	0	0	50	0	0
Obese II	0	0	0	3	0	52	1
Obese III	0	0	0	0	0	0	76

Naive Bayes — Confusion Matrix

Actual ↓ / Predicted →	Normal	Over I	Over II	Obese I	Insufficient Weight	Obese II	Obese III
Normal	27	9	4	0	12	0	4
Over I	7	29	12	4	1	0	0
Over II	8	0	30	20	0	1	2
Obese I	0	8	8	29	0	12	4
Insufficient Weight	2	3	0	0	49	0	0
Obese II	0	0	0	6	0	48	2
Obese III	0	0	0	0	0	0	76

Random Forest — Confusion Matrix

Actual ↓ / Predicted →	Normal	Over I	Over II	Obese I	Insufficient Weight	Obese II	Obese III
Normal	53	3	0	0	0	0	0
Over I	3	48	2	0	0	0	0
Over II	1	0	58	2	0	0	0
Obese I	0	0	0	61	0	0	0
Insufficient Weight	1	0	0	0	53	0	0
Obese II	0	0	0	0	0	55	1
Obese III	0	0	0	0	0	0	76

Apriori Mining – association rule

In Apriori association, it was run on the Lifestyle cleaned dataset with the full 2087 instances. Every one of the strongest patterns we found pointed to the same result for Obesity_Type_III. None of the other six categories showed up this clearly, meaning people in that group can basically be spotted just from their daily habits alone. The table below shows a few examples(rule) along with how often each shows up (support) and how reliable it is (confidence).

Rule	Support	Confidence
Gender=Female family_history_with_overweight=yes FAVC=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes SCC=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FAVC=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes SCC=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes SMOKE=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FAVC=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes SMOKE=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes SMOKE=no SCC=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FAVC=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes SMOKE=no SCC=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' CAEC=Sometimes CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FAVC=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' SMOKE=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99
Gender=Female family_history_with_overweight=yes FCVC='(2.995893-inf)' NCP='(2.999673-3.000487]' SMOKE=no SCC=no CALC=Sometimes MTRANS=Public_Transportation ==> NObeyesdad=Obesity_Type_III	15.4%	0.99

From the table, we see that there are several attributes that are repeatedly occurring across every example. For instance, family history with overweight, high calorie food, and public transportation. For every example the support is 15.4% which shows that these combinations of habits come close to fully identifying individuals in the category of Obesity_Type_III, with a confidence level of 0.99. It seems that most rules include high vegetable consumption, which is counterintuitive and may be tied to overall calorie intake rather than vegetable consumption itself. But overall, these examples show the association between lifestyle habits and the last stage of obesity which is the most severe.

Conclusion

To conclude, from this project examined if an individual's obesity category can be predicted from eating habits and lifestyle attributes alone rather than from BMI. If combination of habits tends to occur more often with higher obesity levels. Using lifestyle attributes (no weight/height) , Random Forest had a 88% accuracy then an increasing accuracy with All (including weight/height) attributes of a 96% , showing the lifestyles attributes still had a significant contribution to an individual's obesity. The Apriori analysis, every example showed a confidence.99 to the Obesity_Type_III category using specific combination of categories. Overall, it shows how eating habits and physical activities are very useful in indicating obesity risk but it cannot directly declare the outcome.

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